Imperial College
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Lecture 7

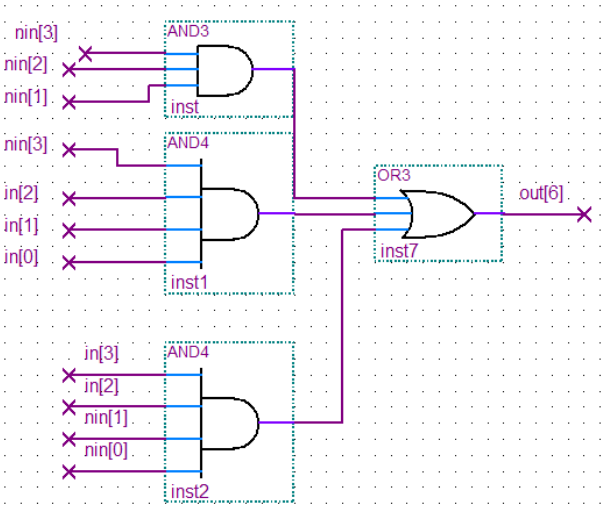
Digital Design with FPGA

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How to describe/specify digital circuits?

Schematic diagram & gates



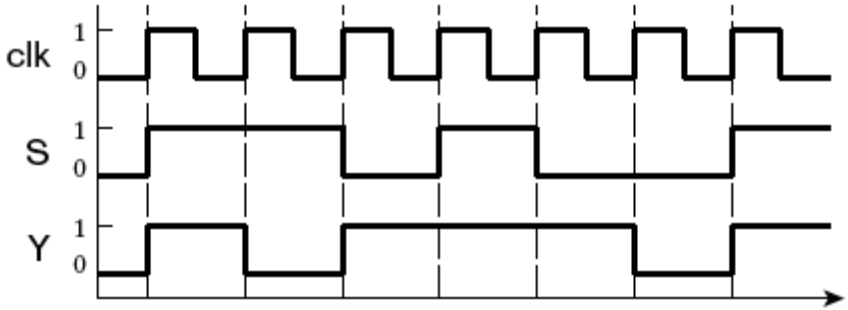
Boolean equation

$$out6 = /in3*/in2*/in1 + in3*in2*/in1*/in0 + /in3*in2*in1*in0$$

Truth table

in[3..0]	out[6:0]
0000	1000000
0001	1111001
0010	0100100
0011	0110000
0100	0011001
0101	0010010
0110	0000010
0111	1111000
1000	0000000
1001	0010000

Timing Diagram



Hardware Description Language (HDL)

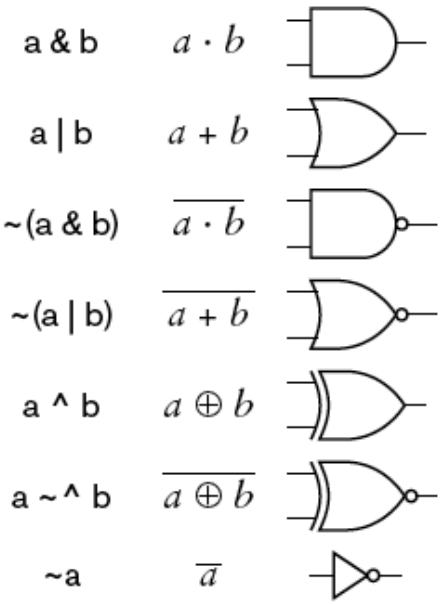
3-to-1 MUX

```
module mux32three(i0,i1,i2,sel,out);
input [31:0] i0,i1,i2;
input [1:0] sel;
output [31:0] out;
reg [31:0] out;

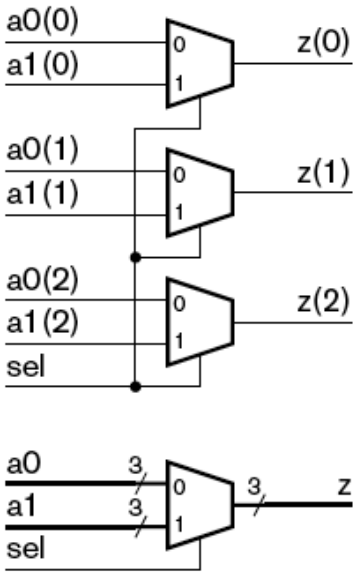
always @ (i0 or i1 or i2 or sel)
begin
  case (sel)
    2'b00: out = i0;
    2'b01: out = i1;
    2'b10: out = i2;
    default: out = 32'bx;
  endcase
end
endmodule
```

Basic digital building blocks

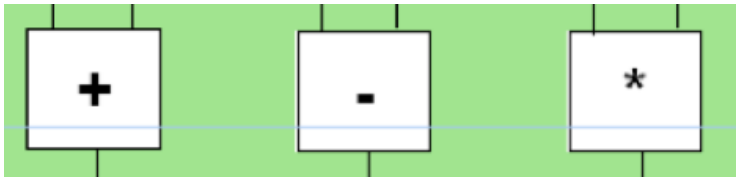
Primitive Logic Gates



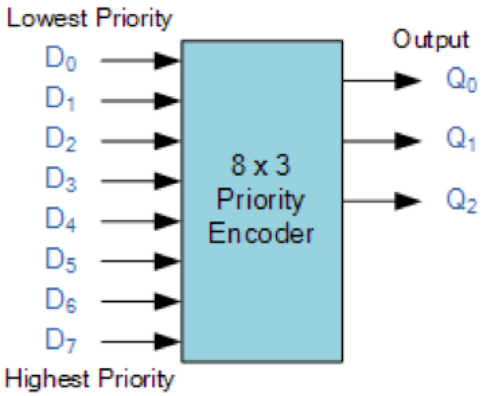
Multiplexers



Arithmetic circuits

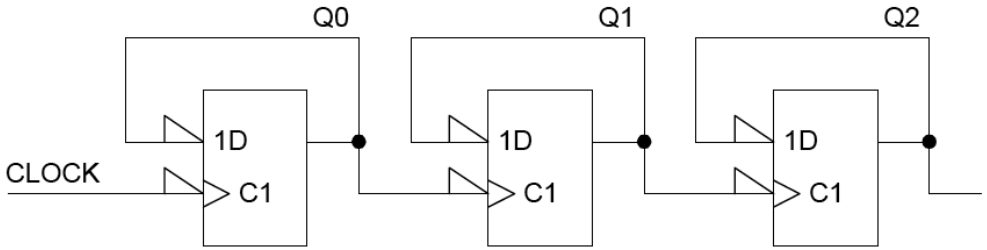


Encoders

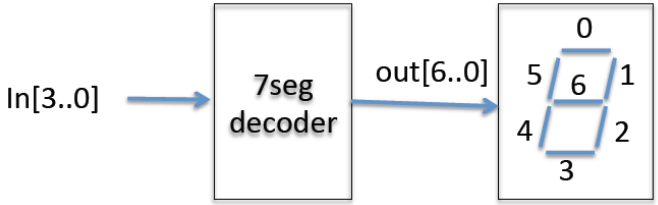


Inputs								Outputs		
D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	Q_2	Q_1	Q_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	x	0	0	1
0	0	0	0	0	1	x	x	0	1	0
0	0	0	0	1	x	x	x	0	1	1
0	0	0	1	x	x	x	x	1	0	0
0	0	1	x	x	x	x	x	1	0	1
0	1	x	x	x	x	x	x	1	1	0
1	x	x	x	x	x	x	x	1	1	1

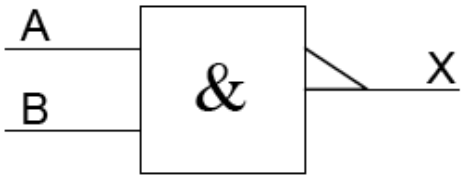
Flipflops and Registers

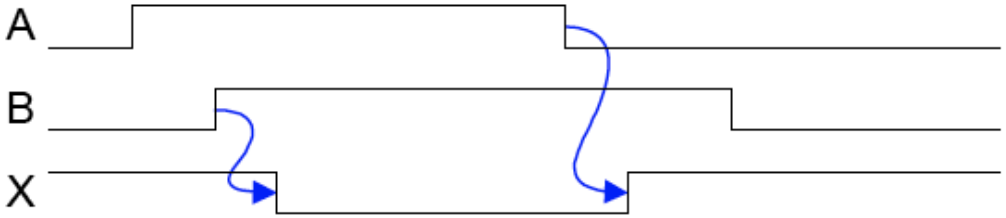


Decoders



Cause & Effect





Input B going high causes X to go low

Input A going low causes X to go high

Propagation Delay:

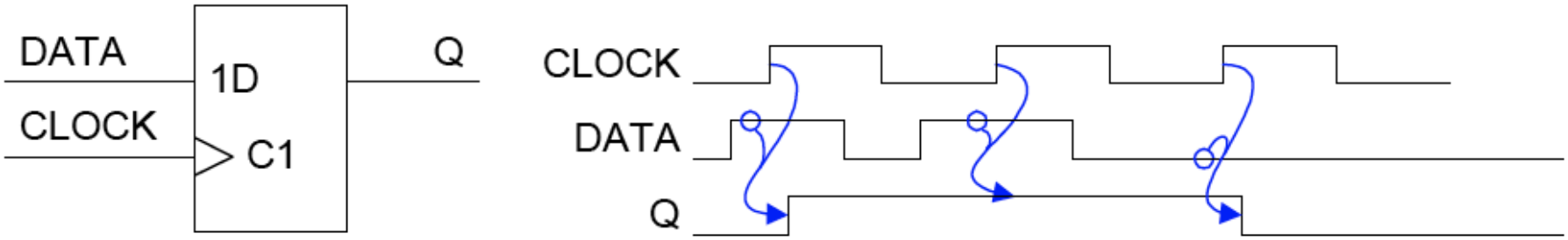
The time delay between a cause (an input changing) and its effect (an output changing), assuming output load capacitance of 30pF.

Example: 74AC00: Advanced CMOS 2-input NAND gate

	min	typ	max	
A↑ to X↓ (t_{PHL})	1.5	4.5	6.5	ns
A↓ to X↑ (t_{PLH})	1.5	6.0	8.0	ns

t_{PHL} and t_{PLH} refer to the direction that the output changes: high-to-low or low-to-high.

D-Flipflop (1)



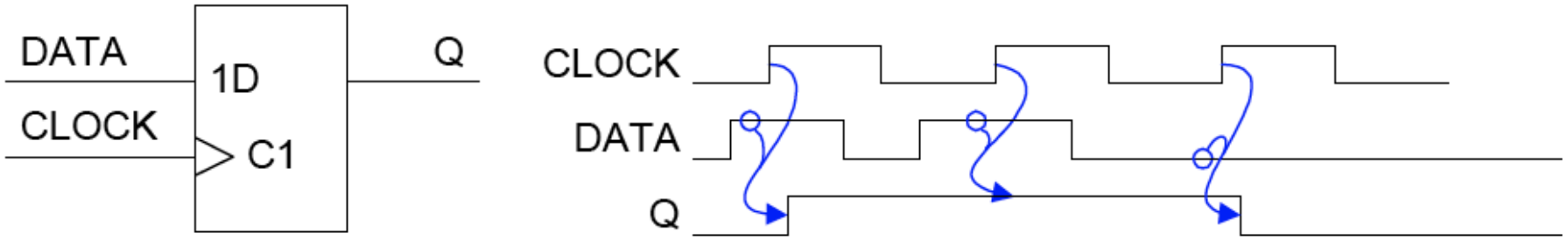
Notation:

>	input effect happens on the rising edge
C1	C ⇒ Clock input, 1 ⇒ This input *is* input number 1.
1D	D ⇒ Data input, 1 ⇒ This input *is controlled by* input number 1.

The meaning of a number depends on its position:

A number *after* a letter is used to identify a particular input.
A number *before* a letter means that this input is controlled by one of the other inputs.

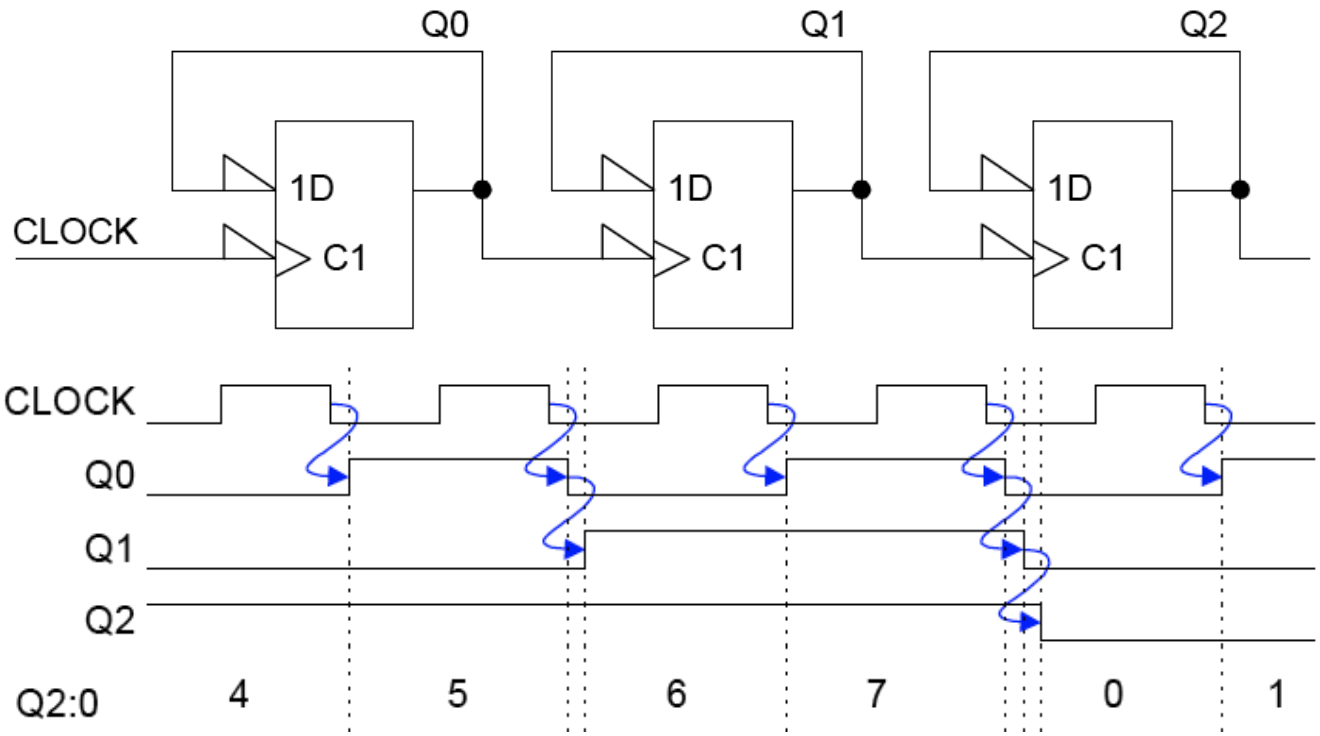
D-Flipflop (2)



Cause and Effect:

- CLOCK↑ causes Q to change after a short delay. This is the *only* time Q ever changes.
- The value of D *just before* CLOCK↑ is the new Q.
- Propagation delay CLOCK↑ to Q is typically 1 ns.
- Propagation delay DATA to Q **does not make sense** since DATA changing does not cause Q to change.

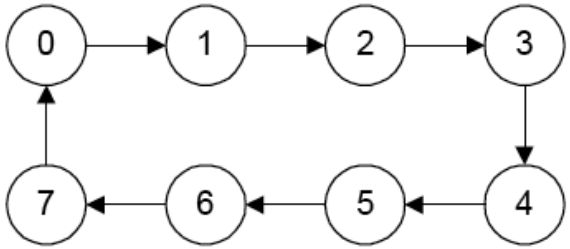
Ripple Counter



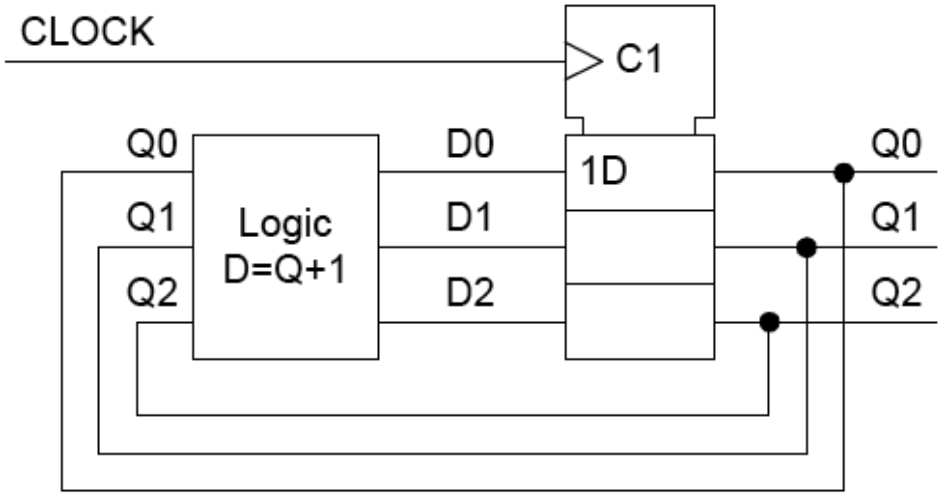
- Notice inverters on the CLOCK and DATA inputs
- Least significant bit of a number is always labelled 0

Propagation Delay: CLOCK↓ to Q2 = 3 x 1ns = 3ns

State Diagram
(not including transient states):

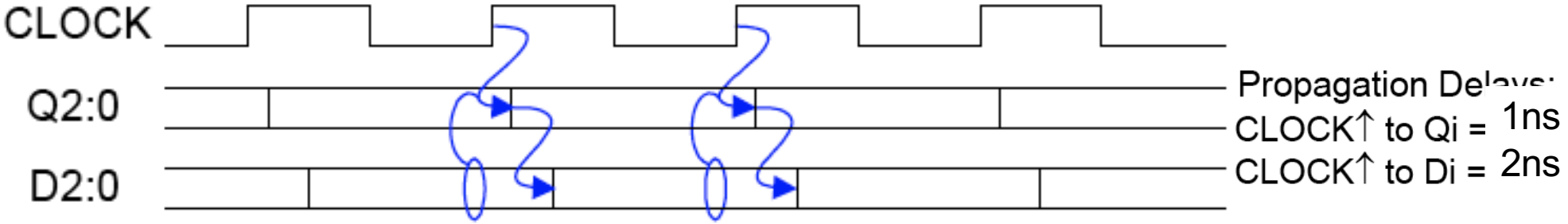


Synchronous Counter



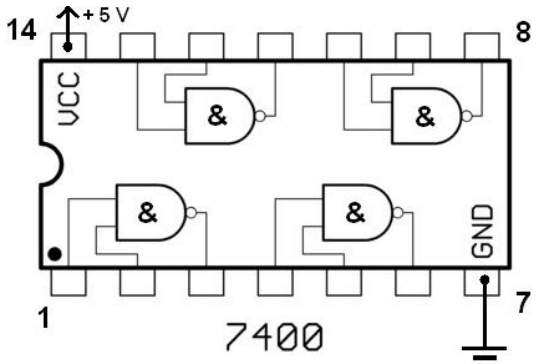
- A *register* is a bunch of flipflops with the same CLOCK.
- The individual flipflops are rectangles stacked on top of each other. Only the top one is labelled.
- All shared signals (e.g. the CLOCK input) go to the notched *common control block* at the top of the stack.

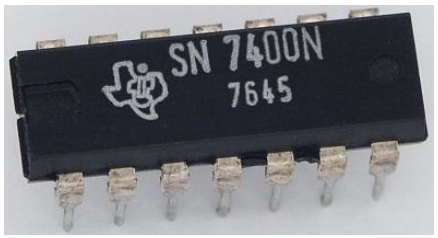
The logic block must add 1 onto the current value of the counter, Q, to generate the next value of the counter, D. Suppose it has a propagation delay of 1ns.



All flipflops change state within a fraction of nanosecond of each other.

Old ways of implementing digital circuits

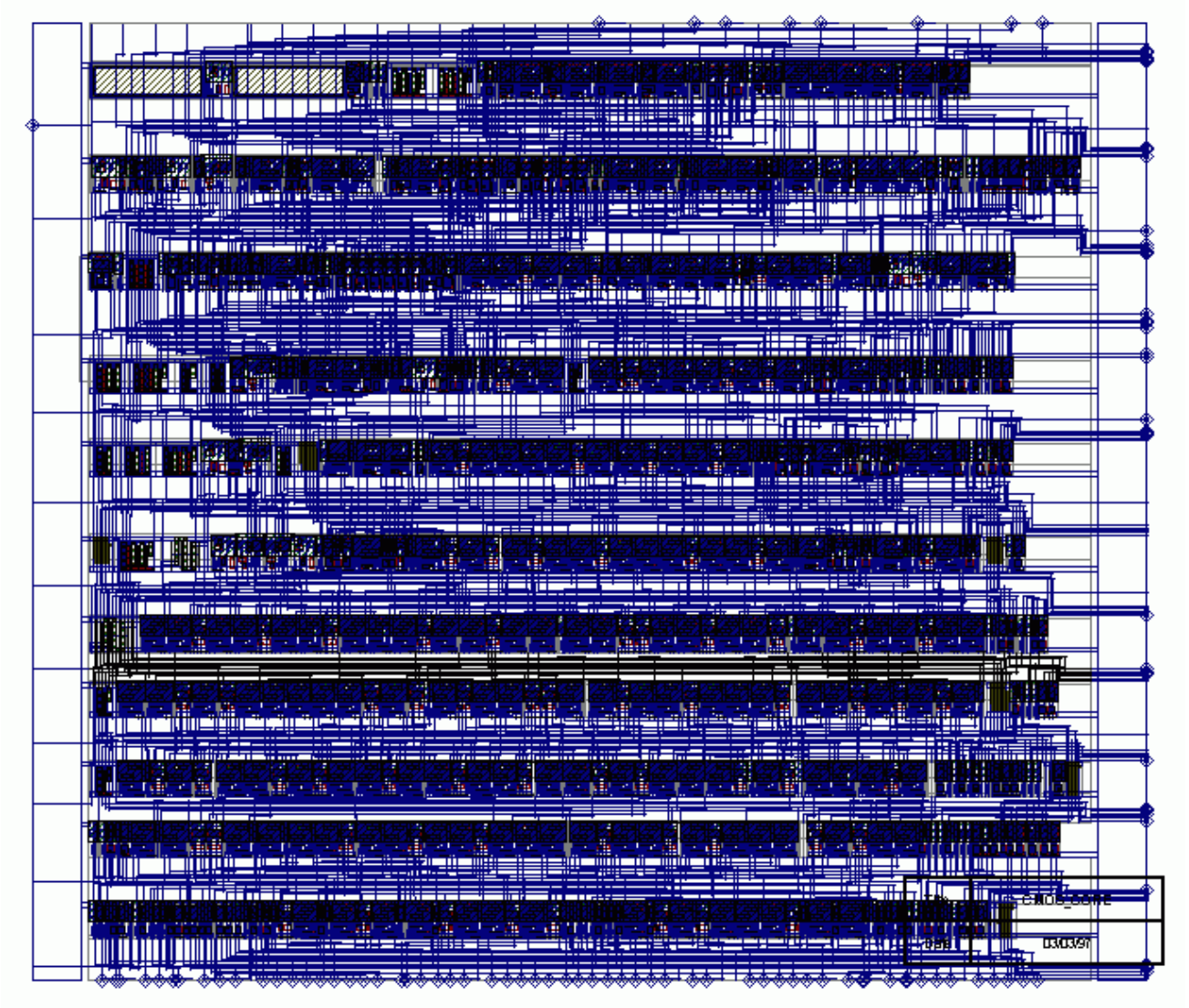




- Discrete logic – based on gates or small packages containing small digital building blocks (at most a 1-bit adder)
- De Morgan's theorem – theoretically we only need 2-input NAND or NOR gates to build anything
- Tedious, expensive, slow, prone to wiring errors



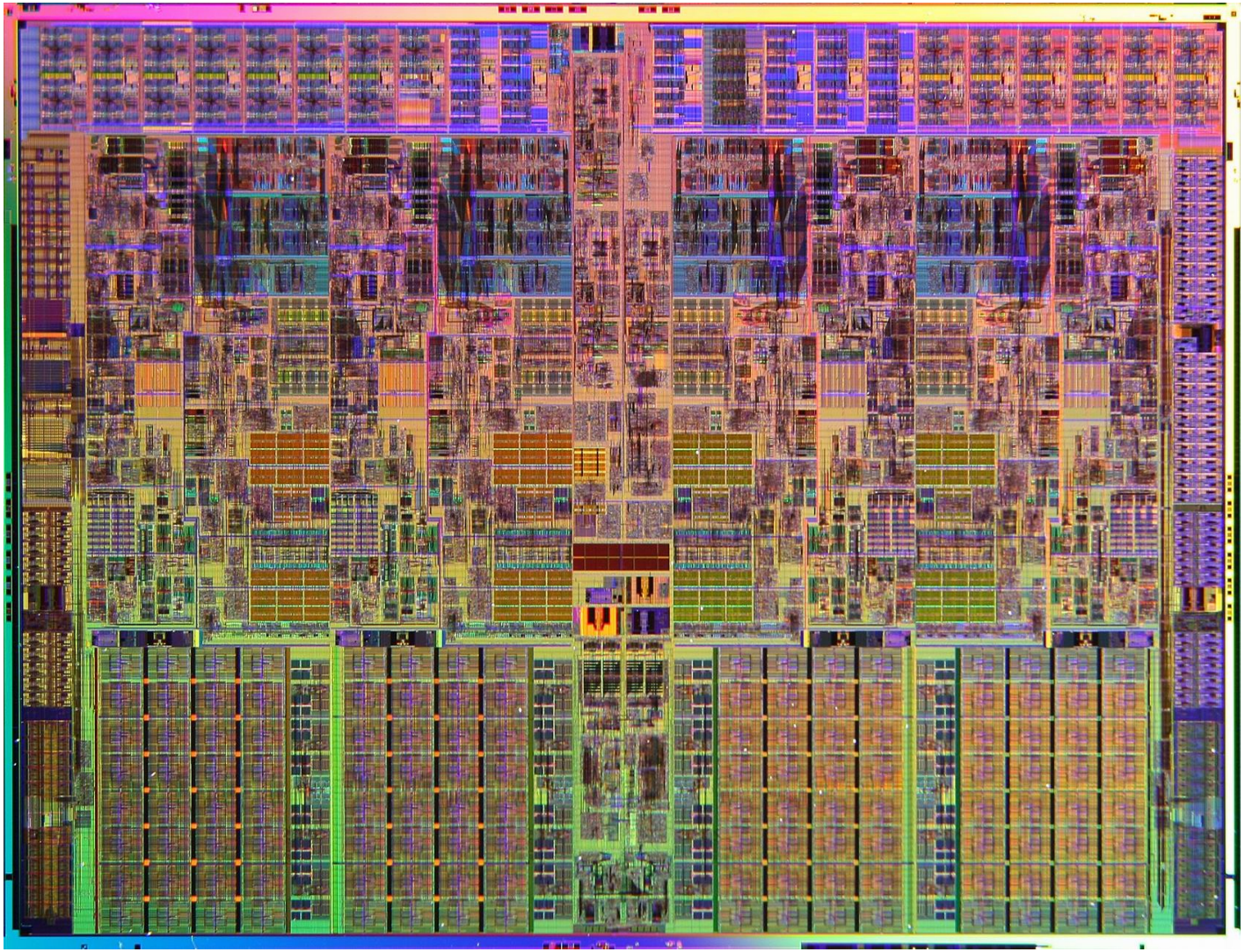
Early integrated circuits based on gate arrays



- Rows of gates – often identical in structure
- Connected to form customer specific circuits
- Can be full-custom (i.e. completely fabricated from scratch for a given design)
- Can be semi-custom (i.e. customisation on the metal layers only)
- Once fabricated, the design is fixed

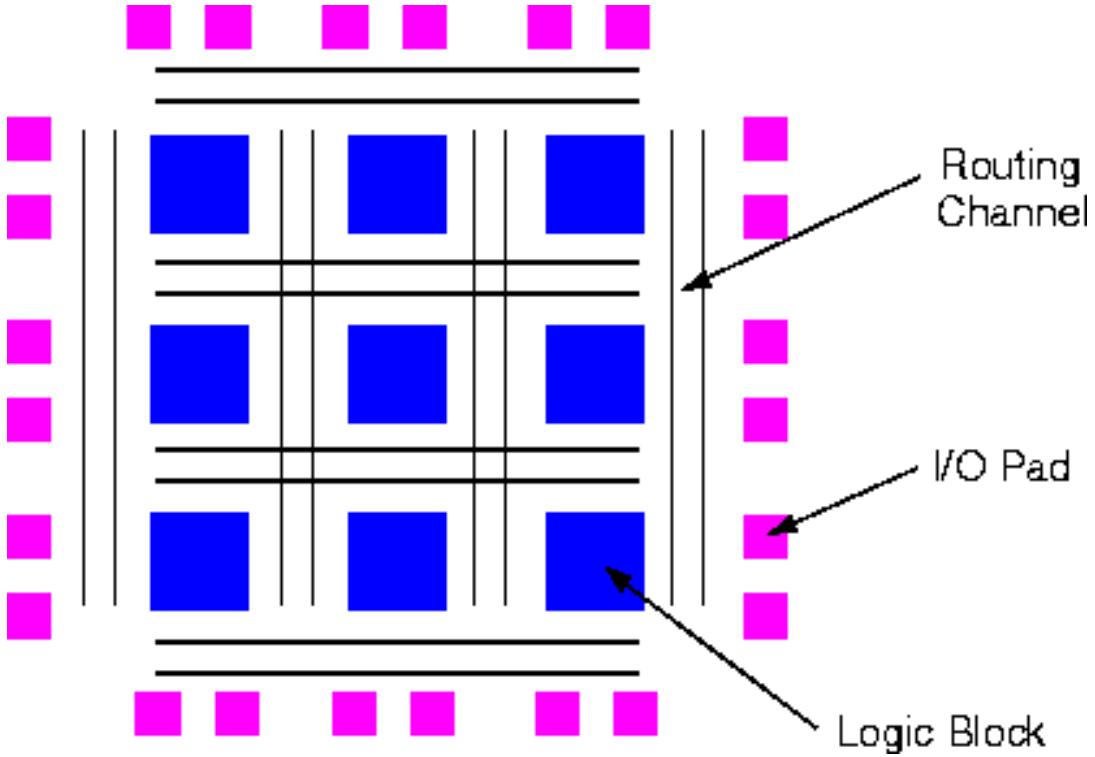
Modern digital design – full custom IC

- Intel Core i7
- > ¾ billion trans.
- Very expensive to design
- Very expensive to manufacture
- Not viable unless the market is very large



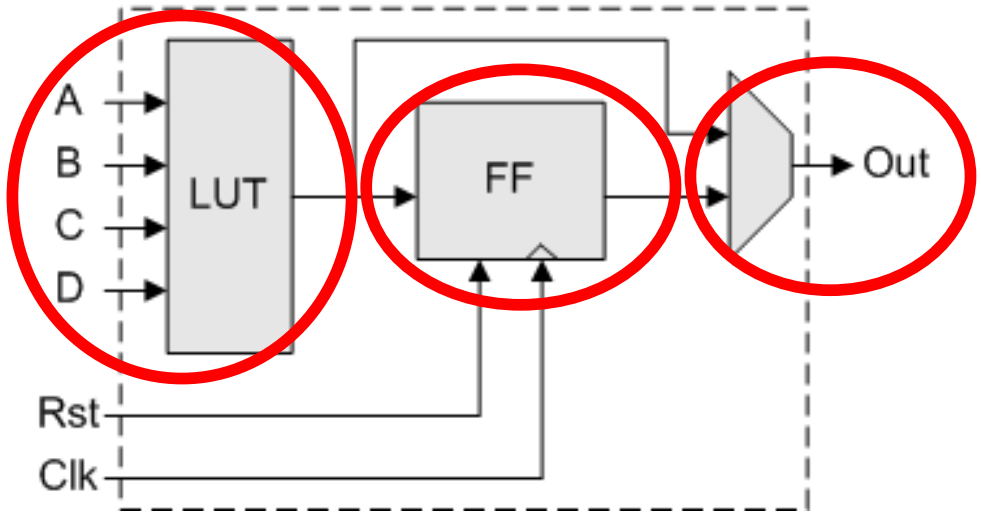
Field Programmable Gate Arrays (FPGAs)

- Combining idea from Programmable Logic Devices (PLDs – Yr 1 Lecture 8) and gate arrays
- First introduced by Xilinx in 1985
- Arrays of logic blocks (to implement logic functions)
- Lots of programmable wiring in routing channels
- Very flexible I/O interfacing logic core to outside world
- Two dominant FPGA makers:
  - Xilinx and Altera
- Other specialist makers e.g. Actel and Lattice Logic

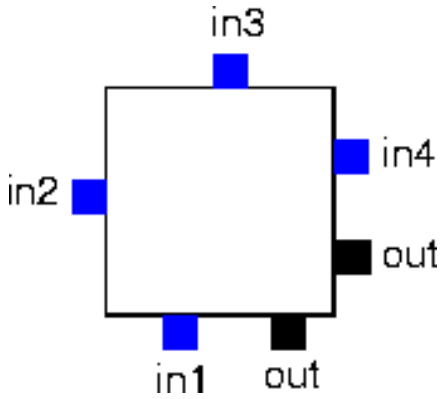


Configurable Logic Block (or Logic Element)

- Based around Look-up Tables (LUTs), most common with 4-inputs
- Optional D-flipflop at the output of the LUT
- 4-input LUT can implement ANY 4-input Boolean equation (truth-table)
- Special circuits for cascading logic blocks (e.g. carry-chain of a binary adder)



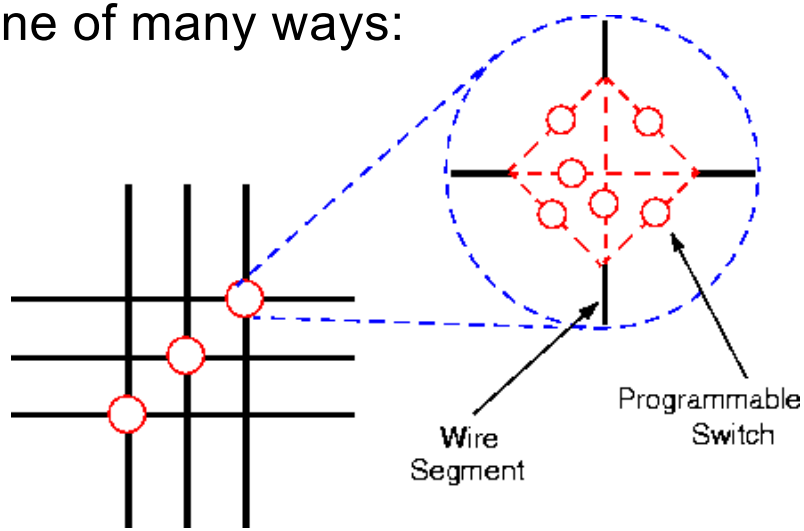
Logic Element (LE)

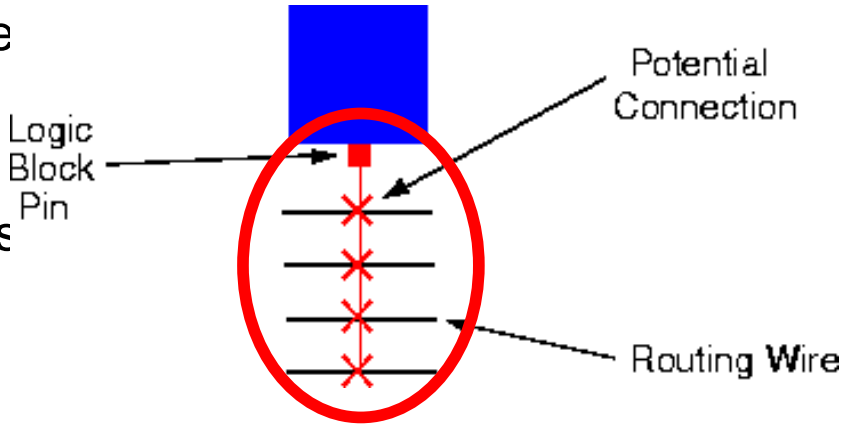


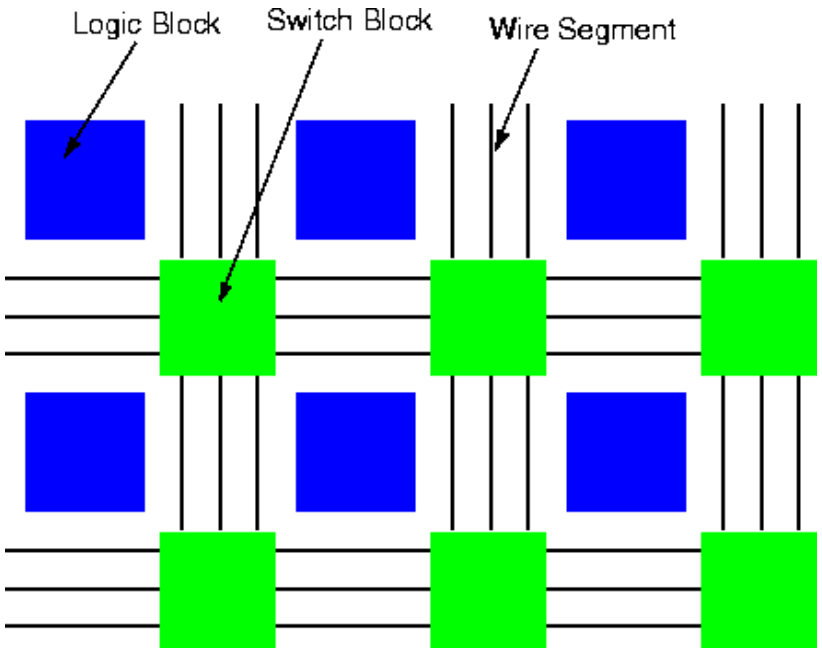
- Each logic block has pins located for easy access:

Programmable Routing

- Between rows and columns of logic blocks are wiring channels
- These are programmable – a logic block pin can be connected to one of many wiring tracks through a programmable switch
- Xilinx FPGAs have dedicated switch block circuits for routing (more flexible)
- Each wire segment can be connected in one of many ways:

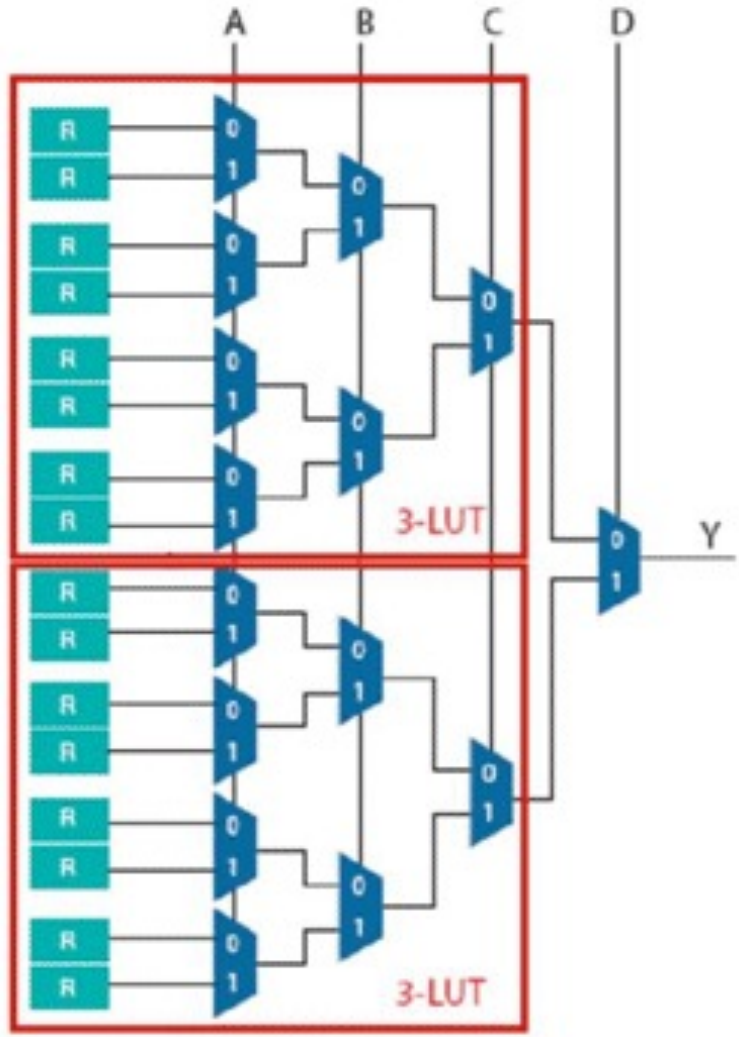






The Idea of Configuring the FPGA

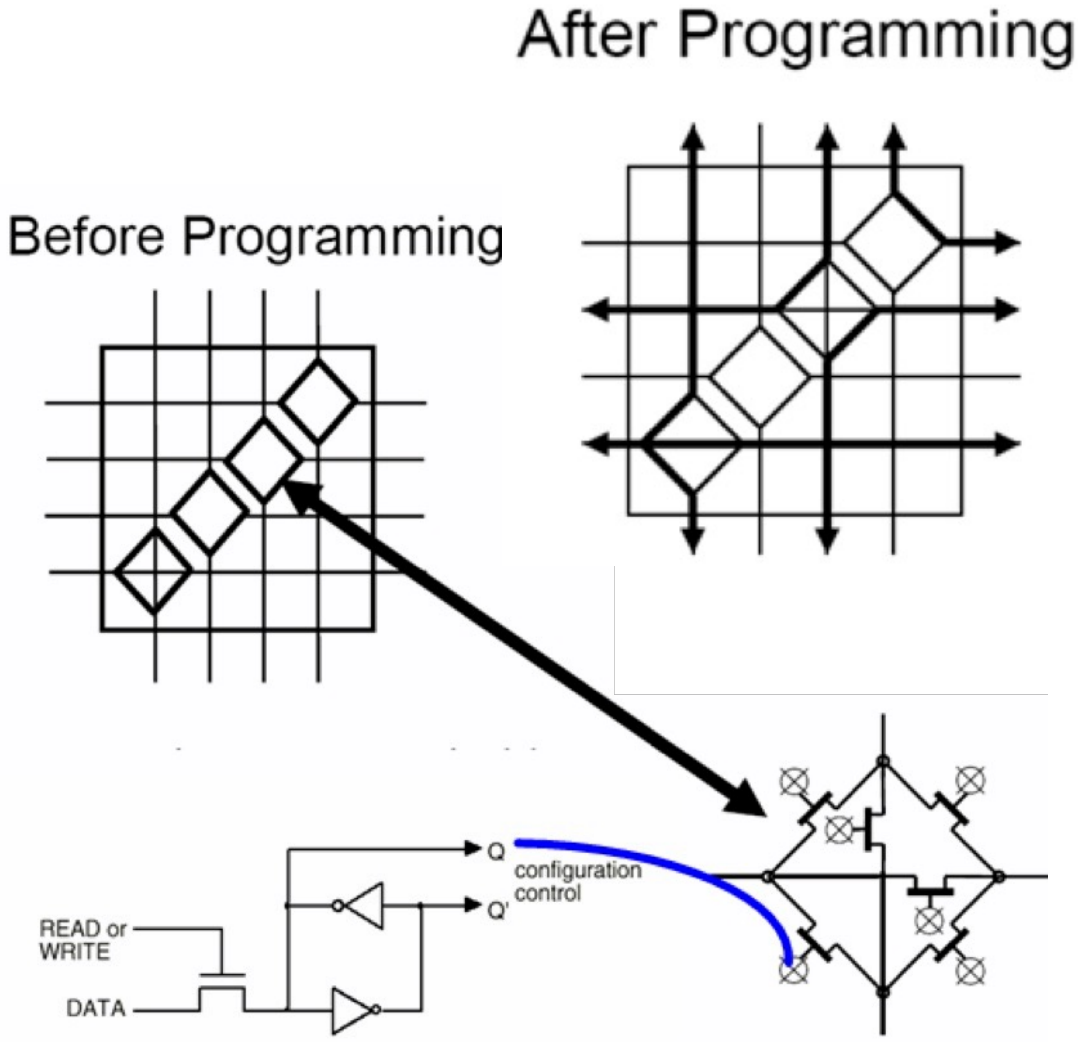
- Programming an FPGA is NOT the same as programming a microprocessor
- We download a **BITSTREAM** (not a program) to an FPGA
- Programming an FPGA is known as **CONFIGURATION**
- All LUTs are configured using the BITSTREAM so that they contain the correct values to implement the Boolean logic
- Shown here is a typical implementation of a 4-LUT circuit
  - ABCD are the FOUR inputs
  - There are four levels of 2-to-1 multiplexer circuits
  - The 16-inputs to the mux tree determine the Boolean function to be implemented as in a truth-table
  - These 16 binary values are stored in registers (DFF)
  - Configuration = setting the 16 registers to 1 or 0



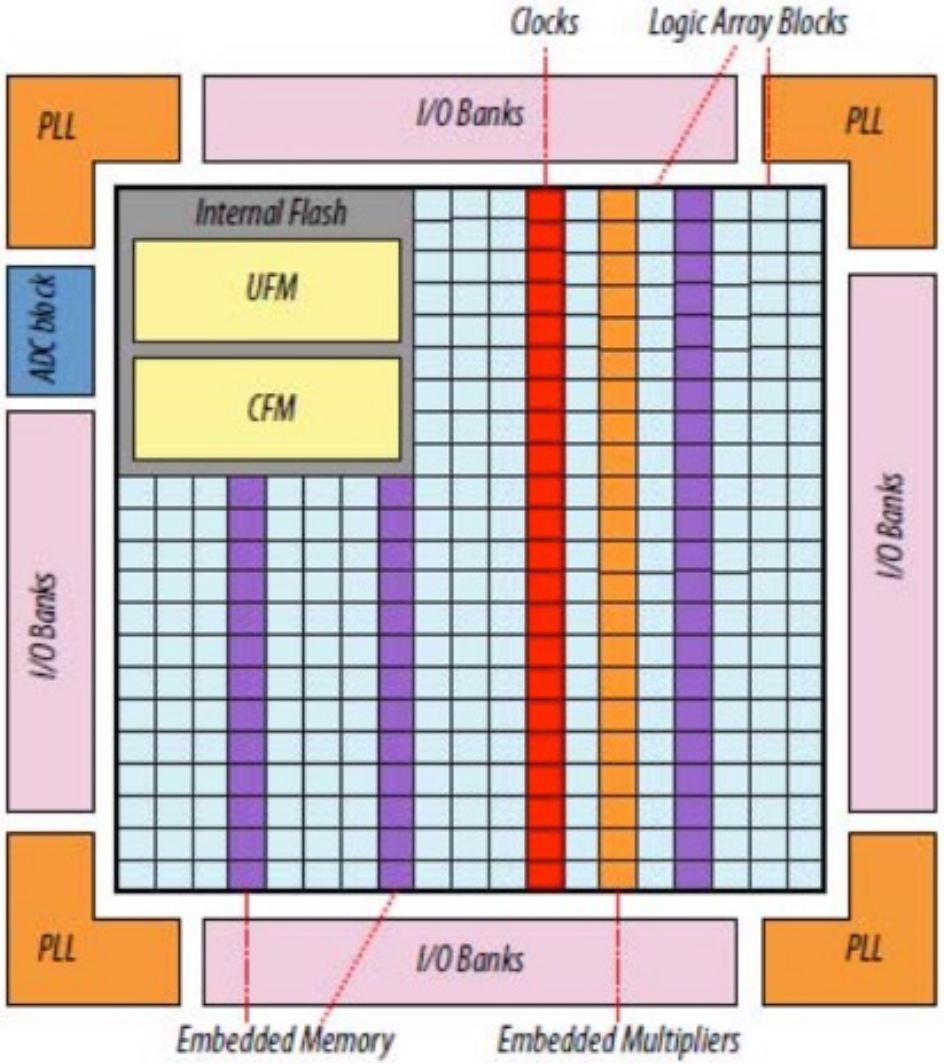
Configuring the routing in an FPGA

- At each interconnect site, there is a transistor switch which is default OFF (not conducting)
- Each switch is controlled by the output of a 1-bit configuration register
- Configuring the routing is simply to put a '1' or '0' in this register to control the routing switches

- Bitstream is either stored on local flash memory or download via a computer
- Configuration happens on power-up



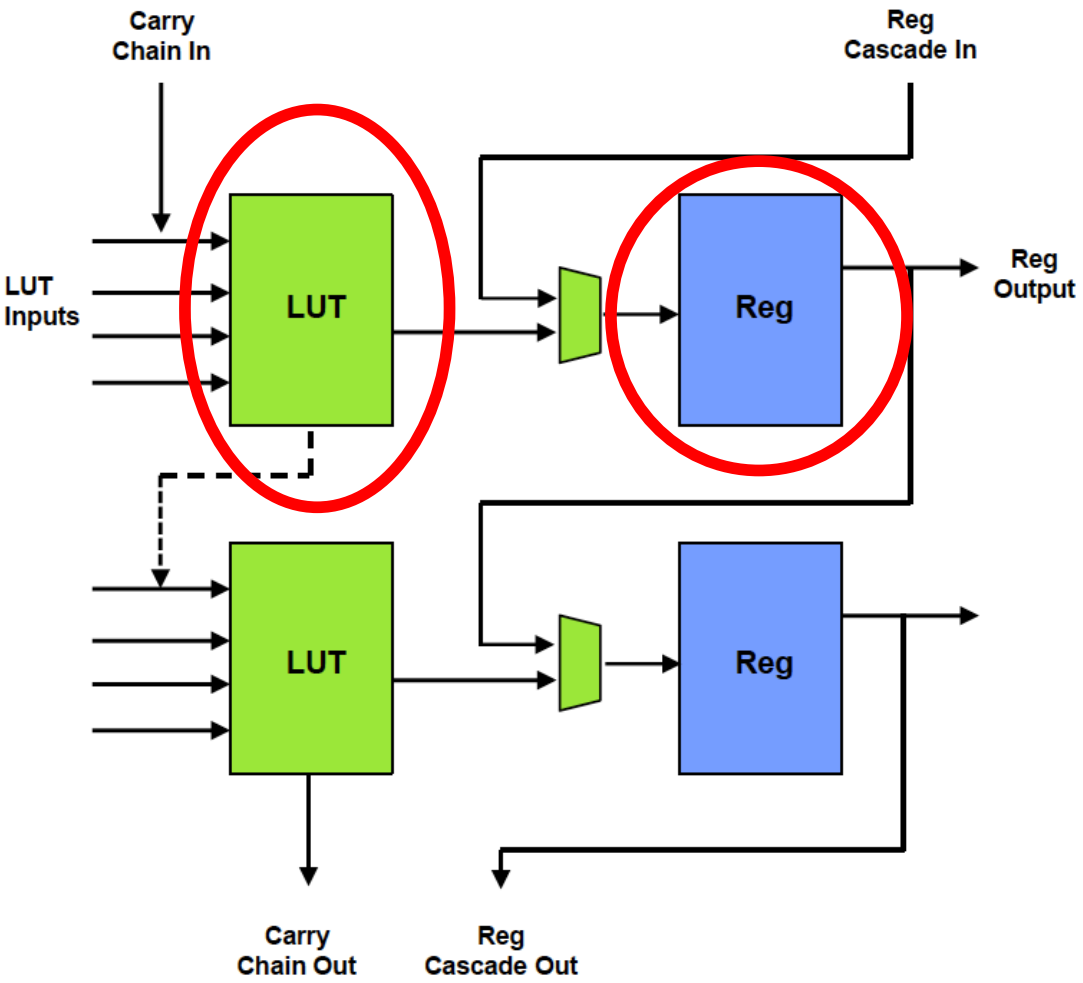
Max 10 FPGA – chip level view



- Intel/Altera's Max 10 FPGA
- Low cost family

- Inside FPGA, it has:
  - Logic Array Blocks (LABs) containing Logic Elements (LEs)
  - Embedded Memory Blocks
  - Embedded Multiplier Blocks
  - Clock Blocks
  - Internal Flash (program/data)
  - Configurable I/O circuits
  - Phase-locked Loops (PLL)
  - Analog-to-Digital converter Block

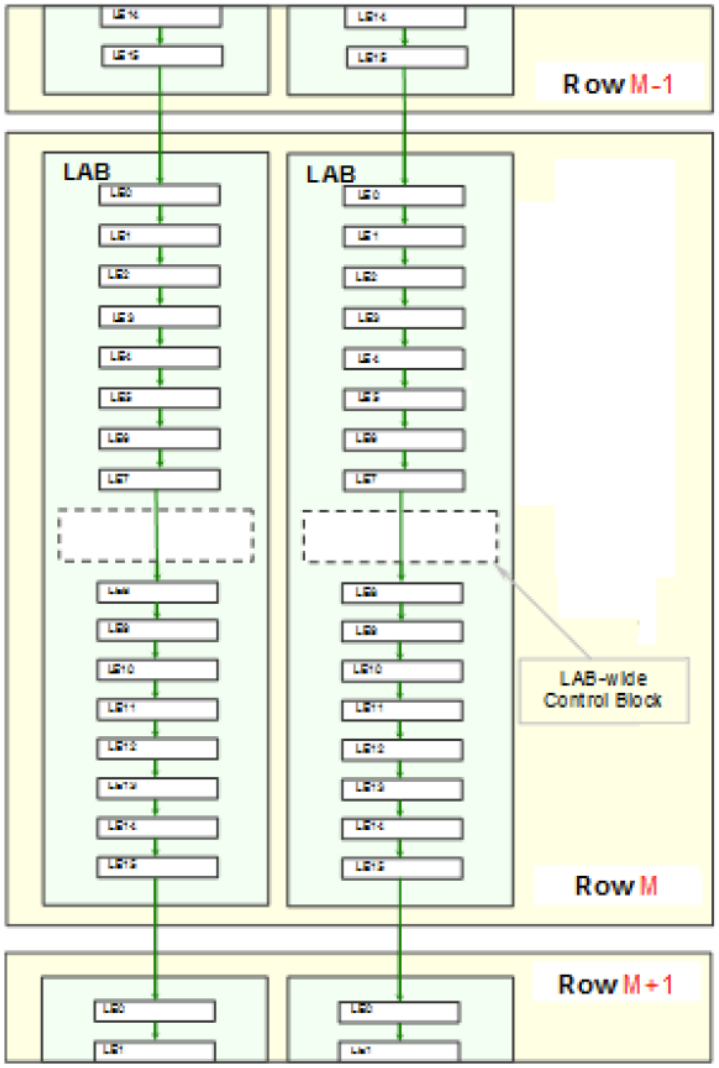
Max 10 FPGA Logic Element



- Intel/Altera's Max 10 FPGA
- Device is **10M50DAF484C7G**
- 50,000 Logic Elements (LEs)
- Classical 4-LUT with D-FF

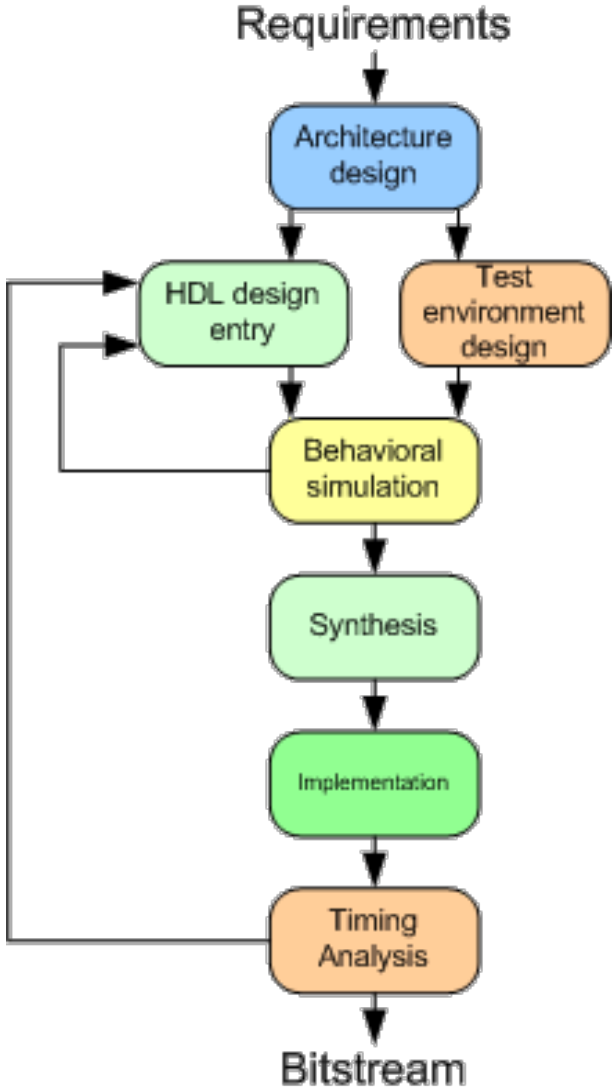
- Two dedicated paths between LEs:
  - Carry chain
  - Register cascade chain

Max 10 Logic Array Block (LAB)



- Logic Element (LE)
  - 4 LUT
  - Register
  - Output routing logic
- Logic Array Block
  - 16 Logic Elements
  - Local Lab Control
  - Routing inside LAB

Design Tools – Quartus Prime Lite



- Quartus Prime Lite – a comprehensive design tools for Intel/Altera FPGAs
- Special web edition free to download from (need registration):
  - https://fpgasoftware.intel.com/18.0/?edition=lite&platform=windows
  - Features include (see introduction to Quartus II):
    - design entry
    - compilation from Hardware Description Languages (HDL)
    - synthesis
    - simulation
    - timing analysis
    - power analysis
    - project management

DE10-Lite FPGA Board

